

Copper and iron ions accelerate the prion-like propagation of α -synuclein: A vicious cycle in Parkinson's disease

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ABSTRACT

Protein fibrils drive the onset and progression of many diseases in a prion-like manner, *i.e.* they transcellularly propagate through the extracellular space to health cells to initiate toxic aggregation as seeds. The conversion of native α -synuclein into filamentous aggregates in Lewy bodies is a hallmark of Parkinson's disease (PD). Copper and iron ions accumulate in PD brains, however, whether they influence the prion-like propagation of α -synuclein remain unclear. Here, we reported that copper/iron ions accelerate prion-like propagation of α -synuclein fibrils by promoting cellular internalization of α -synuclein fibrils, intracellular α -synuclein aggregation and the subsequent release of mature fibrils to the extracellular space to induce further propagation. Mechanistically, copper/iron ions enhanced α -synuclein fibrils internalization was mediated by negatively charged membrane heparan sulfate proteoglycans (HSPGs). α -Synuclein fibrils formed in the presence of copper/iron ions were more cytotoxic, causing increased ROS production, cell apoptosis, and shortened the lifespan of a *C. elegans* PD model overexpressing human α -synuclein. Notably, these deleterious effects were ameliorated by two clinically used chelators, triethylenetetramine and deferiprone. Together, our results suggest a new role for heavy metal ions, *e.g.* copper and iron, in the pathogenesis of PD through accelerating prion-like propagation of α -synuclein fibrils.

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1. Introduction

Parkinson's disease (PD) is a common age-associated neurodegenerative disease [1–3], which features progressive death of dopaminergic neurons and conversion of soluble α -synuclein (α -syn) protein into filamentous aggregates in Lewy bodies [4,5]. α -Syn is a 140-amino acid protein that plays roles in memory, recognition, and regulation of dopamine and presynaptic vesicle trafficking [6–8]. Studies have shown that α -syn fibrils could spread along neuron network [9–11], or even spread from gut to brain via the vagus nerve [12,13], in a prion-like manner, *i.e.* the fibrillar proteins could induce aggregation of native proteins [14–16]. Such prion-like propagation among cells is associated with the release of α -syn fibrils into the extracellular space [17], once internalized by recipient cells, these aggregates may function as seeds to

trigger aggregation of native intracellular α -syn, which initiates and/or exacerbates PD [18]. Although this phenomenon is well-described, the mechanisms influencing such prion-like fibrils propagation remain largely unclear.

Accumulation of metal ions has been reported in PD brains. For examples, iron concentration is elevated nearly three-fold in the substantia nigra of PD patients [19]; copper accumulates and positively correlates with the increasing percentage of PD in aging brains [20,21]; the copper concentration in cerebrospinal fluid increases from 0.2 μ M in normal people to 0.5 μ M in PD patients [22]. However, it remains unknown whether copper and iron ions play roles in the transcellular propagation of α -syn.

Here, we investigated how copper/iron ions affect the propagation of α -syn. Their effects on extra- and intra-cellular α -syn aggregation, internalization of α -syn fibrils by recipient cells, and the propagation of α -syn fibrils, were assessed. Mechanistically, how copper/iron ions affect inter-cellular α -syn propagation, and the cell membrane molecules involved in propagation, were identified. Therapeutic interventions

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using chelating agents were evaluated *in vitro* and in a *C. elegans* PD model overexpressing human α -syn.

2. Results and discussion

2.1. Copper and iron ions accelerate the propagation of α -syn aggregation

Prion-like propagation of α -syn is described as extracellular fibrils internalized by recipient cells as seeds to induce intracellular aggregation of native α -syn. Methods have been developed to indicate intracellular α -syn aggregation [23]. Here, a fluorescence resonance energy transfer (FRET) based approach was applied in HEK293 cells expressing α -syn fusions with CFP (α -syn-CFP) and YFP (α -syn-YFP), a FRET-pair that quantitates intracellular aggregation based on fluorescence (Fig. 1A) [24,25].

We investigated the influence of copper/iron ions on α -syn fibril propagation in two models, respectively simulating exogenously added fibrils and physiologically cell-released fibrils. In model one, preformed α -syn fibrils were added to medium as seeds (Fig. 1A, top). Enhanced intracellular aggregation was observed as indicated by increased FRET (Fig. 1B–C). The presence of copper/iron ions concentration-dependently enhanced intracellular α -syn aggregation in the seeded groups (Fig. 1B–C), but showed no obvious effect on seed-free controls (Fig. S1). As a disease highly-related species [26], the effects of metal ions on oligomers should be further explored in future studies.

In model two, a co-culture cell assay was established to simulate transcellular propagation of α -syn fibrils (Fig. 1A, bottom). HEK293 cells expressing a fusion of α -syn with a truncated EGFP (synT), which spontaneously forms fibrils and released to medium (Fig. S1C–D), was designated as the fibril-originated cells [27,28]. Fluorescence of α -syn-CFP/YFP expressing recipient cells indicates intracellular α -syn aggregation. After co-culturing fibril-originated cells with an equal number of recipient cells for 48 h, significantly increased intracellular α -syn-CFP/YFP aggregation was observed in co-cultured cells, indicating the success of this transcellular propagation model (Fig. 1D–E). Further dose-dependently enhanced α -syn aggregation was observed after addition of copper/iron ions, suggesting these metal ions exacerbate transcellular propagation (Fig. 1D–E). To exclude the effects of copper and iron ions on the fibril-originated cells, a medium exchanging experiment was performed. By using medium of cells expressing synT to culture cells expressing α -syn-CFP/YFP, in the presence or absence of copper and iron ions (Fig. S2A), similar results were obtained compared to those of the co-culture experiments (Fig. S2B–C).

Inter-cellular propagation of fibrils involved multi-steps, including internalization of fibrils by recipient cells, seeded fibril formation, and subsequently the release of mature fibrils, we next investigated how copper/iron ions influence different stages.

2.2. Copper and iron ions facilitate seeded aggregation of α -syn

Thioflavin-T (ThT) fluorescence assay was used to monitor aggregation of α -syn in the presence or absence of copper/iron ions. Consistent with a previous report [29], without pre-existing seeds, copper/iron ions by themselves accelerated α -syn aggregation (Fig. S3), possibly due to the high protein concentration and shaking conditions. Moreover, preformed fibrils were more prone to initiate α -syn aggregation than copper/iron ions. In the presence of preformed α -syn fibrils (seeds, 5%), the aggregation lag time of α -syn accelerated from 3.8 d to 4.2–4.6 h (Fig. 2 & Table 1), shorter than the aggregation lag time of seed-free α -syn with copper/iron ions (0.5 and 3.1 d, respectively, Fig. S3). The presence of copper/iron ions further concentration-dependently enhanced seeded aggregation of α -syn, as indicated by the reduced lag times and increased fluorescence intensities (Fig. 2A–B & Table 1). Consistent observations were also found by the turbidity assays (Fig. S4A), and by measurement of soluble protein (Fig. S4B).

Morphologically, α -syn aggregates exhibited different fibrillar shapes depending on treatments (Fig. 2C).

Secondary structural transition of α -syn solution during aggregation was determined using far-UV circular dichroism (CD) spectroscopy. At the beginning of incubation, a 200 nm negative peak was detected in all groups, suggesting dominant random coil structures. During incubation, this negative peak was gradually uplifted, indicating a conversion from random coils to β -structures (Figs. 2D & S5A–C). Structural transition for seeded α -syn started after 8 h of incubation, which was accelerated to 4 h in the presence of equal molar of copper or iron ions, indicating accelerated transition (Fig. 2D).

The secondary structure of α -syn aggregates was analyzed using Fourier Transform Infrared Spectroscopy (FTIR). Aggregates formed with α -syn seeds were composed of β -sheet structure (31.0%), β -turn (26.8%) and random coil (42.2%) (Figs. 2E & S5D). While for seeded α -syn co-incubated with copper/iron ions (Cu-fibrils & Fe-fibrils), the content of β -sheet increased to 38.9% and 39.9%, and β -turn structure decreased to 11.3% and 16.5%, respectively (Figs. 2E & S5D). These results suggest that copper/iron ions cause α -syn fibrils of different secondary structures. Conformations of α -syn fibrils have been reported to affect the propagation of fibrils [30–32], propagation-prone fibrils are suggested to have β -sheet rich structures [33–35]. G51D α -syn fibrils, with higher β -sheet contents than wild-type α -syn fibrils, induce formation of α -syn inclusions at a faster rate, more widespread pathology and severer progressive nigral neuronal loss than wild-type α -syn fibrils [36]. Consistently, Cu-/Fe-fibrils as seeds accelerated the aggregation of α -syn at faster rates than wild-type α -syn fibril seeds (Fig. S6).

2.3. Copper/iron ions promote the internalization of fibrils

To investigate whether copper/iron ions affect the internalization of α -syn fibrils, SH-SY5Y (human neuroblastoma) and N2a (mouse neuroblastoma) cells, were treated with 100 nM preformed fluorescein isothiocyanate (FITC) labeled α -syn (α -syn-FITC) fibrils, which were pretreated with or without iron and copper ions for 12 h. In control cells not exposed to α -syn-FITC fibrils, no green FITC fluorescence was detected (Fig. S7). While in α -syn-FITC fibrils treated SH-SY5Y cells, 46.7% cells were FITC fluorescence positive, indicating internalization of α -syn fibrils (Fig. 3A–B); the presence of 100 μ M copper or iron ions promoted internalization of α -syn fibrils (71.3% and 61.0%, Fig. 3A–B). Similar results were obtained with N2a cells (Fig. 3C–D).

2.4. Copper/iron-promoted α -syn fibrils internalization requires heparan sulfate proteoglycans (HSPGs)

We further investigated how copper/iron ions promote the internalization of fibrils. HSPGs is a linear polysaccharide on extracellular membrane, which regulates a range of biological activities like angiogenesis, blood coagulation and tumor metastasis [37], and has also been previously implicated to involved in amyloid internalization [38,39]. Hence, we tested whether HSPGs mediate copper/iron-promoted internalization of α -syn fibrils.

Heparin, an isomer of heparan sulfate, was used to competitively inhibit the interaction between HSPGs and α -syn fibrils [40]. In the presence of heparin, copper/iron ions no longer promoted the internalization of α -syn fibrils and the resulting intracellular aggregation (Figs. 4A–C & S8A–B). We further knocked down Ext1, a key biosynthesis enzyme for HSPG production [41], by shRNAs (Fig. S8C). Copper/iron ions enhanced internalization of α -syn fibrils was largely blocked by Ext1 knockdown (Fig. 4A–B). These results indicate that HSPGs mediates copper/iron ions promoted internalization of α -syn fibrils.

Next, we explored how copper/iron ions affect α -syn-HSPGs interaction. It has been reported that copper/iron ions bind to amyloidogenic proteins like α -syn and amyloid- β , co-existing in the formed fibrils, and modifying fibril properties such as surface electrical charge [42–44]. Since HSPGs are net negatively charged while copper/iron ions are

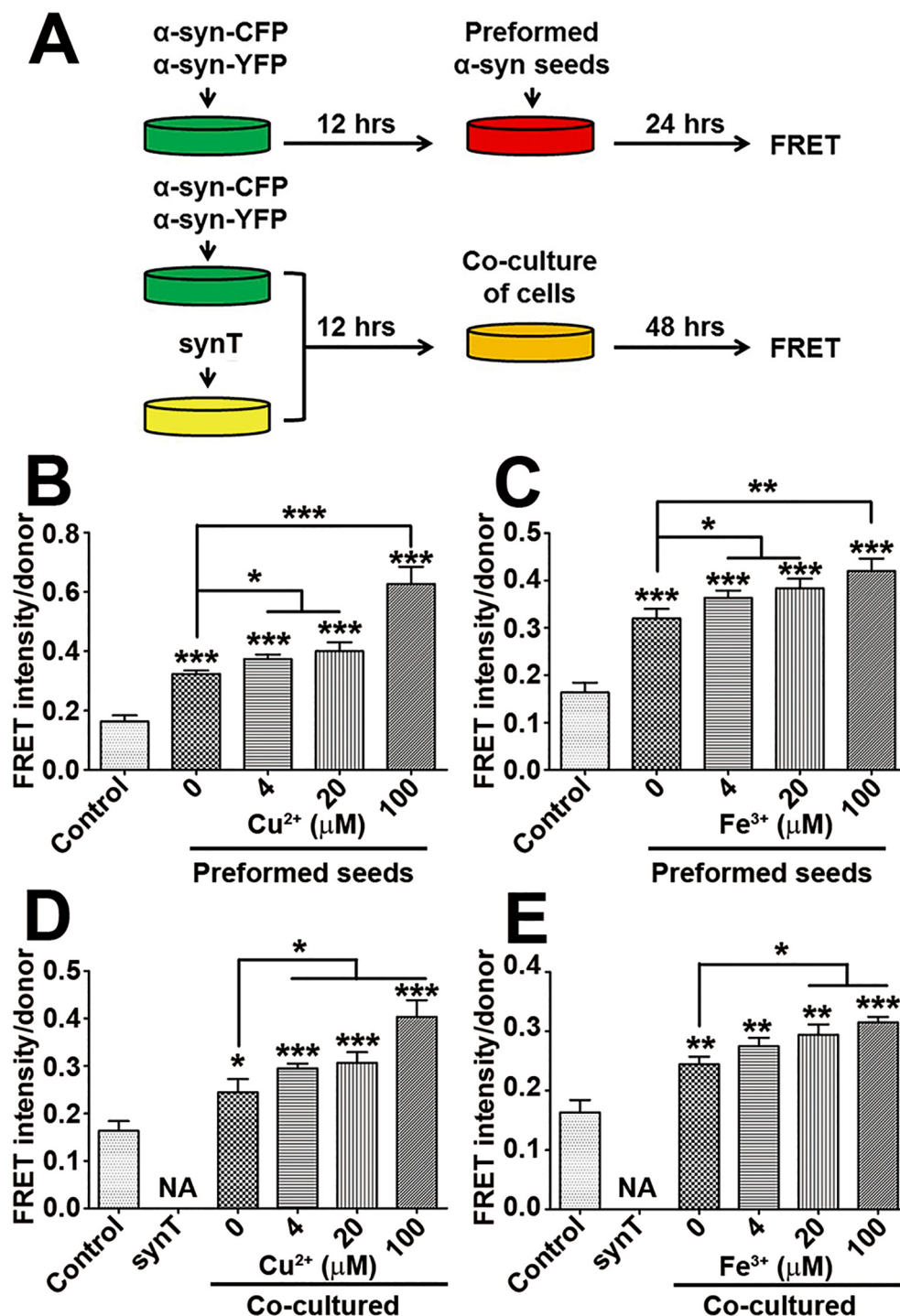


Fig. 1. Copper and iron ions accelerate propagation of α -syn aggregation as assessed by FRET. (A) Schematic representation of two seeded aggregation models of α -syn-CFP/YFP to produce FRET fluorescence. Model 1, HEK293 cells co-transfected with α -syn-CFP/YFP were treated with 500 nM α -syn fibrils in the presence or absence of copper/iron ions for 24 h before FRET measurement (top). Model 2, HEK293 cells expressing α -syn-CFP/YFP were co-cultured with an equivalent number of cells expressing synT for 48 h to monitor transcellular propagation of α -syn protein aggregation (bottom). (B–C) Copper (B) and iron (C) ions accelerated preformed fibril seeded intracellular aggregation in a cell-based FRET assay. (D–E) Copper (D) and iron (E) ions accelerated transcellular propagation. The control was cells expressing α -syn-CFP/YFP. NA, data not available for cells expressing synT because they do not produce fluorescence. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

positively charged, we speculate that charge-charge interaction may account for copper/iron enhanced binding of α -syn to HSPGs. To test this possibility, we measured the zeta potential and found increased net positive charge of α -syn fibrils treated with copper/iron ions, as well as fibrils formed in the presence of copper/iron ions (Cu-/Fe-fibrils, Fig. 4D). Chemical modification is usually used to change the surface

property of proteins [45]. We further modified mature α -syn fibrils on their surface carboxyl groups with N-(3-Dimethylaminopropyl)-N'-ethylcarbodiimide (EDC), N-hydroxysulfo-succinimide (sulfo-NHS) and ethylamine (Fig. S8D), which increased the net positively charge of α -syn fibrils (Fig. 4E), and enhanced internalization were also found (Fig. 4F–G).

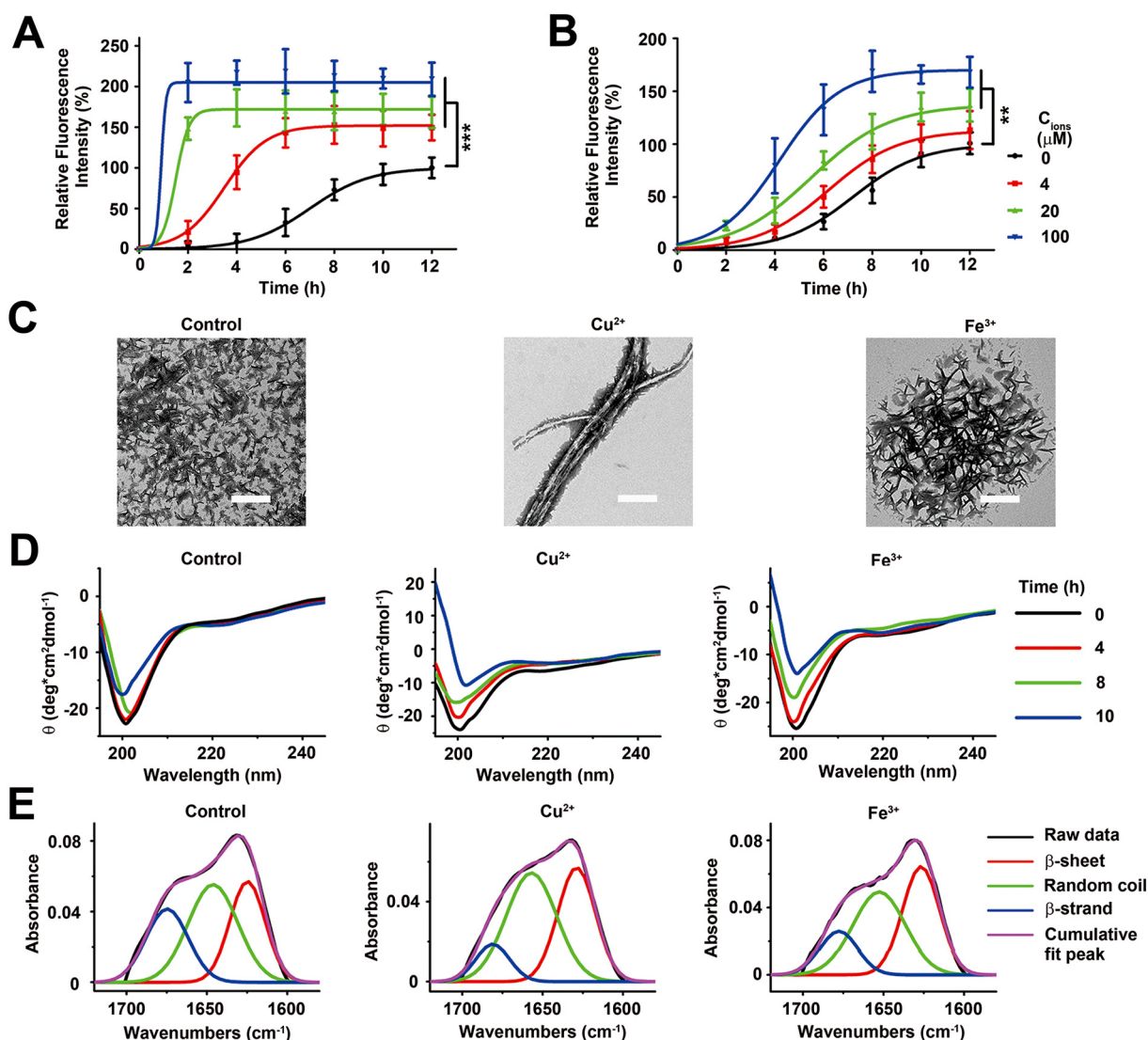


Fig. 2. Copper and iron ions facilitate seeded aggregation of α -syn. (A–B) Seeded (5%) aggregation measured by ThT fluorescence of 100 μ M α -syn in the presence of 4, 20 and 100 μ M copper (A) or iron (B) ions. (C) TEM images of α -syn samples incubated for 12 h; the scale bar represents 200 nm. (D) CD spectra of seeded (5 μ M fibrils) aggregation of 100 μ M α -syn with or without 100 μ M copper or iron ions. (E) FTIR spectra and the fitted curves of fibrils of α -syn, or α -syn with 100 μ M copper ions or iron ions. Each experiment was performed in triplicate and repeated at least three times.

2.5. Copper/iron ions increase the release of mature fibrils

A critical step for amyloid propagation is the release of mature fibrils to extracellular space [17]. To investigate whether copper/iron ions affect the release of intracellular fibrils, we transfected HEK293 cells

with Flag- α -syn, collected the medium, lyophilized [46,47], and probed with anti-Flag antibody (Fig. 5A). Western blots indicated monomers and high-order Flag- α -syn oligomers in the medium, after being induced by α -syn or synT fibrils (Fig. 5B). Moreover, the presence of copper/iron ions enhanced release of Flag- α -syn aggregates (Fig. 5B).

Cell death contributes to the release of intracellular fibrils to extracellular space [48]. Accordingly, we analyzed the cytotoxicity of the aggregates formed in the presence of copper/iron ions (Cu-/Fe-fibrils) by MTT assays. The results demonstrated that mature α -syn fibrils showed no significant cytotoxicity (Figs. 5C & S9A), which is consistent with previous reports [49,50]. In contrast, Cu- and Fe-fibrils reduced SH-SY5Y cell viability by 13.7% and 10.3%, respectively (Fig. 5C); and reduced N2a cell viability by 21.3% and 15.1%, respectively (Fig. S9A), demonstrating enhanced cytotoxicity.

Increasing evidence has established that fibrils may induce cell damage through mitochondrial dysfunction and impaired protein degradation [51,52]. Reactive oxygen species (ROS) are associated with mitochondrial dysfunction and related apoptosis [53,54]. ROS levels and apoptosis-related molecules in cells treated with different α -syn fibrils were detected (Figs. 5D–E & S9B–C). Compared to cells treated

Table 1
Amyloid fibril formation of seeded α -syn^a.

Metal ions	Concentration (μ M)	Lag time (h)	Maximum intensity (%)
Cu^{2+}	0	4.53 ± 0.97	100.00 ± 12.62^b
	4	1.78 ± 0.43	149.59 ± 15.75^b
	20	0.96 ± 0.24	168.29 ± 19.15^b
	100	0.71 ± 0.22	208.94 ± 20.45^b
Fe^{3+}	0	4.37 ± 0.82	100.00 ± 10.29^c
	4	3.30 ± 0.78	113.39 ± 17.89^c
	20	2.34 ± 0.51	136.92 ± 15.50^c
	100	1.77 ± 0.37	167.91 ± 14.59^c

^a α -syn concentration was 100 μ M with 5 μ M seeds.

^b The maximum intensity of the group with 0 μ M Cu^{2+} was set as 100%.

^c The maximum intensity of the group with 0 μ M Fe^{3+} was set as 100%.

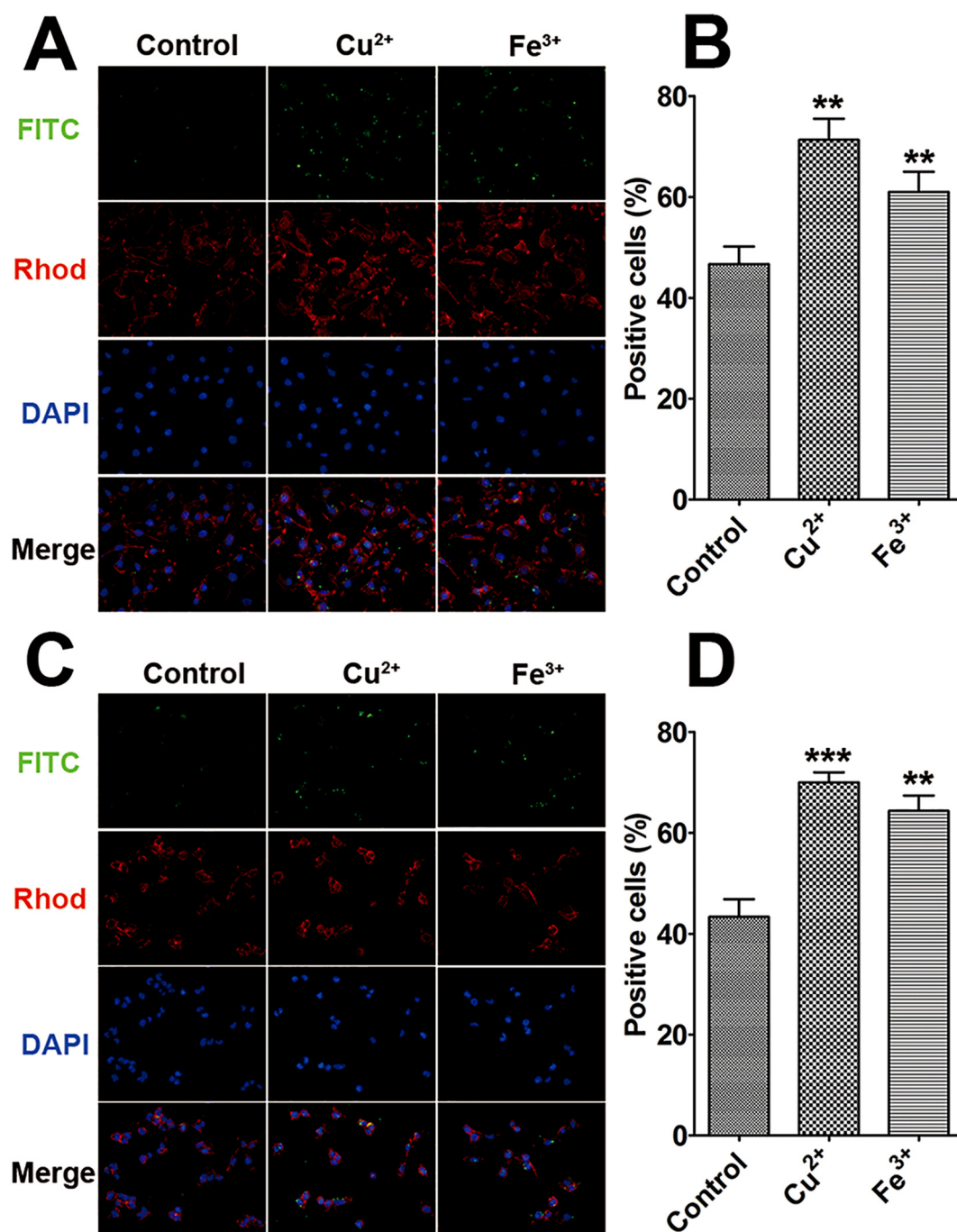


Fig. 3. Copper and iron ions promote internalization of α -syn fibrils. (A–D) SH-SY5Y (A) and N2a (C) cells treated with 100 nM α -syn-FITC fibrils with or without copper/iron ions, and analyzed by fluorescence microscopy and percentage of positive SH-SY5Y (B) and N2a (D) cells for α -syn fibril internalization. The control was treated with α -syn-FITC fibrils alone. Green, FITC labeled fibrils; red, rhodamine labeled phalloidin for indicating cytoskeleton; blue, DAPI for staining nucleus. The images were taken at a magnification ratio of 400 $\times$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

with α -syn fibrils, upregulated apoptosis indicators, such as cytochrome c and cleaved caspase-3, were found in Cu-/Fe-fibrils treated groups, suggesting stronger apoptosis induced by Cu-/Fe-fibrils (Figs. 5E & 5C).

2.6. Chelators abolish the deleterious effects of copper/iron ions in vitro and in vivo

Two clinically used drugs, triethylenetetramine (TETA) and deferiprone (DF), were applied to defend the deleterious effects of copper/irons in the propagation of α -syn *in vitro* and in *C. elegans* models.

TETA and DF are used to treat Wilson's disease and thalassemia, respectively, which are caused by over-accumulation of copper/iron in human body [55,56]. TETA or DF showed no effect on seeded aggregation of α -syn in the absence of copper/iron ions, while they dose-dependently inhibited copper/iron-facilitated seeded aggregation (Fig. 6A & Table 2). Moreover, TETA or DF by themselves showed no detectable effect on α -syn fibril internalization (Fig. S10); however, they reduced the metal associated α -syn fibril internalization in SH-SY5Y (Fig. 6C–D) and N2a cells (Fig. S11). We also investigated the effects of TETA and DF on copper/iron-mediated propagation of α -syn aggregation using FRET.

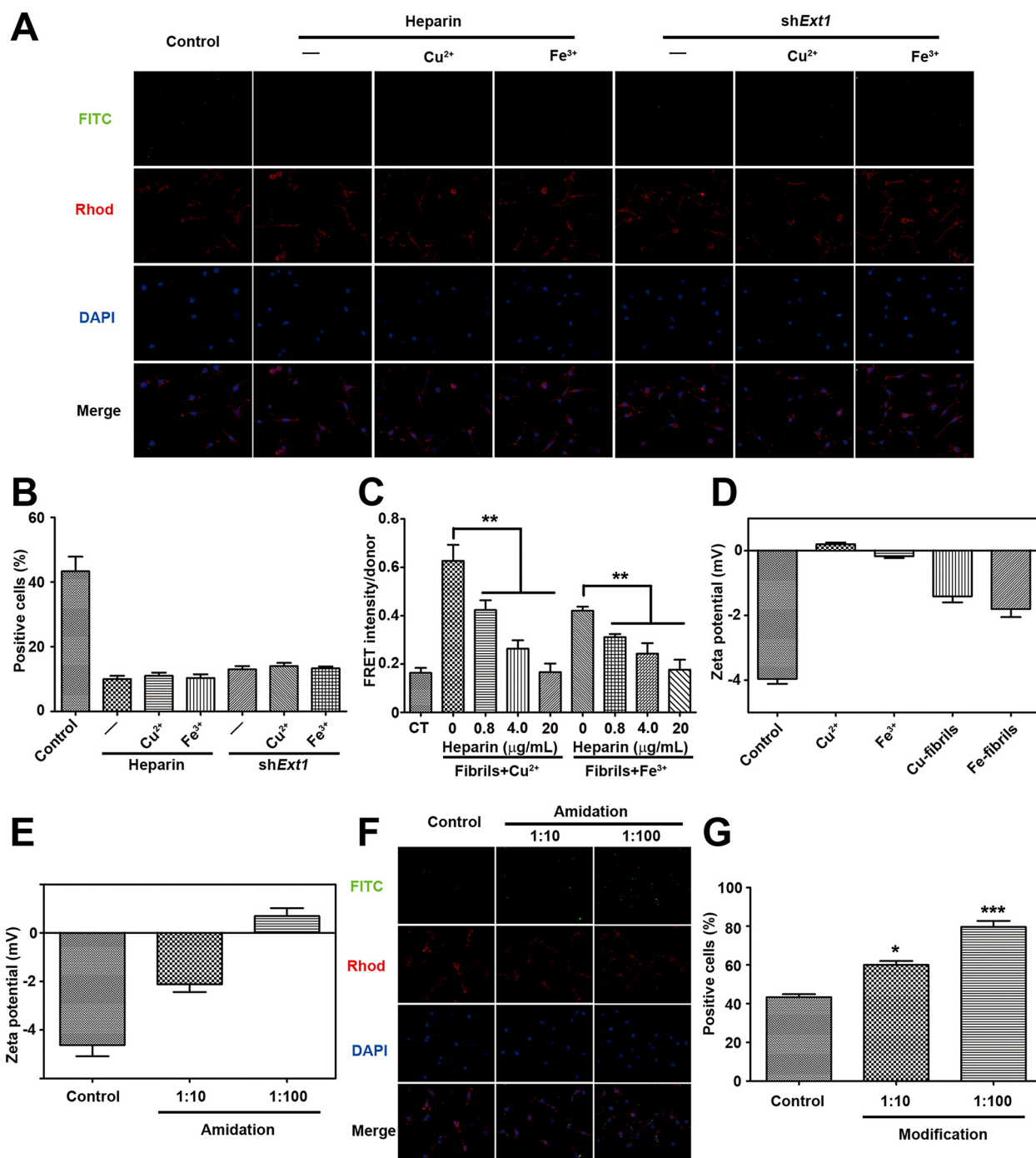


Fig. 4. HSPGs mediates the internalization and propagation of α -syn fibrils. (A–B) Internalization of α -syn-FITC fibrils, in the absence or presence of 100 μ M copper/iron ions, into SH-SY5Y cells was blocked by 20 μ M heparin or knockdown *Ext1*. (C) Heparin dose-dependently inhibited preformed fibrils (500 nM) induced intracellular α -syn-CFP/YFP aggregation in the presence of 100 μ M copper/iron ions. The control was cells expressing α -syn-CFP/YFP without fibril treatment. (D) Zeta potential of 20 μ M α -syn fibrils with or without copper/iron ions, and fibrils formed with copper/iron ions (Cu-/Fe-fibrils). (E–G) The amidation of the carboxyl promoted the net positively charge of α -syn fibrils (E) and fibril internalization (F–G). The amidation was performed with 10 and 100 molar times modification reagents, respectively, including EDC, sulfo-NHS and ethylamine. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

These chelators dose-dependently reduced the accelerating effects of copper/iron ions on α -syn propagation, while neither affected the intracellular α -syn aggregation by itself (Figs. 6E–F & S12).

C. elegans models were applied to study the effects of copper/iron ions *in vivo*. We first determined the dose effect of copper/iron ions on *C. elegans* N2, a strain does not express α -syn (Fig. S13). In transgenic *C. elegans* NL5901, a human α -syn overexpressing PD model, compared to untreated group, decreased life spans were observed in the presence of copper/iron ions, which were rescued by TETA/DF treatment (Fig. 7A). Immuno-dot-blot was applied to identify the contents of

oligomeric-, fibril- and total α -syn of *C. elegans* NL5901 lysates. The results suggested that copper/iron ions accelerated the formation of toxic α -syn oligomers and fibrils, while TETA and DF inhibited the formation of toxic α -syn oligomers and fibrils (Fig. 7B).

3. Conclusion

In conclusion, we reveal that copper and iron ions, which accumulate in PD brains [19,22,57], play accelerating roles during the entire prion-like propagation process of α -syn fibrils *via* a vicious cycle

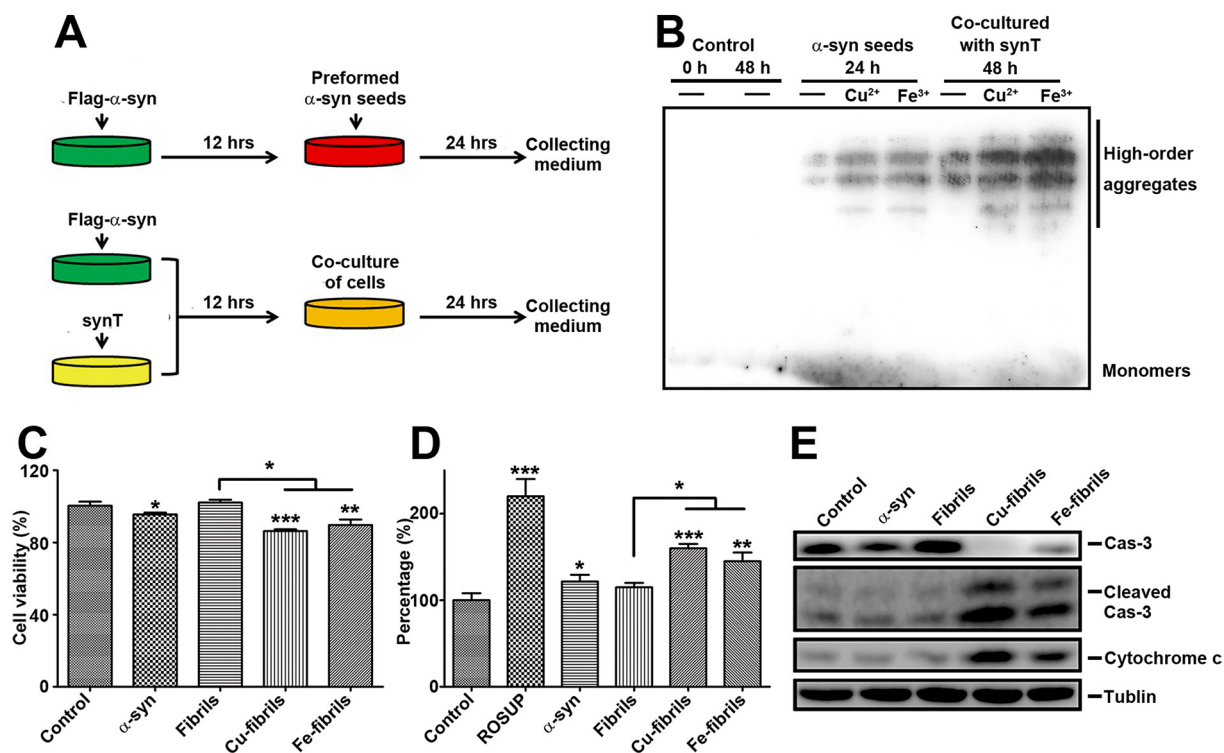


Fig. 5. Copper and iron ions increase release of intracellular aggregates into extracellular space. (A) Schematic representation of experiment to analyze the release of Flag- α -syn aggregates into the medium. HEK293 cells transfected with Flag- α -syn were treated with 500 nM α -syn fibrils for 24 h (top), or co-cultured with an equivalent number of cells expressing synT for 48 h (bottom) in the presence or absence of copper and iron ions. The medium was collected for analysis of Flag- α -syn aggregates using western blots. (B) Western blot results of the Flag- α -syn aggregates in the medium. The control was transfected with synT and treated with α -syn fibrils. (C–D) The cytotoxicity (C) and ROS (D) levels of 10 μ M native α -syn, α -syn fibrils, Cu-fibrils and Fe-fibrils on SH-SY5Y cells. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

(Fig. 7C): i) promoting cellular internalization of fibrils; ii) facilitating internalized fibrils to seed aggregation of native α -syn; iii) enhancing the release of mature fibrils to the extracellular space which induces further propagation. Mechanistically, negatively-charged membrane HSPGs was identified to mediate copper/iron ions promoted fibril internalization *via* charge-charge interaction, and altered secondary structures and morphologies of copper/iron ions induced α -syn fibrils may also contribute to the observed high seeding capacities and cytotoxicity. Notably, clinical used TETA and DF blocked the deleterious effects of copper/iron ions on the toxic propagation of α -syn fibrils, and rescued the life span of a PD C. *elegans* model, indicating a potential therapeutic approach for PD [58]. In addition to α -syn, copper/iron ions have been shown to affect the aggregation or toxicity of other disease-related amyloidogenic proteins, such as amyloid- β , tau and prion [59–61]. Therefore, above conclusions and intervene approaches may also applicable to other protein misfolding diseases [62], which awaits further study.

4. Material and methods

4.1. Materials

α -Synuclein was expressed and purified as previously described [63]. Thioflavin-T, heparin, triethylenetetramine (TETA) and deferiprone (DF) were from Sigma-Aldrich (St Louis, MO). Fluorescein isothiocyanate (FITC) was from Tokyo Chemical (Tokyo, Japan). N-(3-Dimethylaminopropyl)-N'-ethylcarbodiimide (EDC) and N-hydroxysulfosuccinimide (sulfo-NHS) were from Aladdin-Reagents (Shanghai, China). Rhodamine labeled phalloidin was from Yeasen Biotech (Shanghai, China). Antibodies that probe oligomers (A11, #AB9234, Merck Millipore), fibrils (OC, #AB2286, Merck Millipore), α -syn (Syn204, ab3309, Abcam), caspase-3 (9665, CST, Cell Signaling

Technology, Danvers, MA), cleaved caspase-3 (9661, CST), cytochrome c (4280, CST), Flag (F1804, Sigma-Aldrich), β -actin (#GB13001, Google Biology, Wuhan, China), and tubulin (AT819, Beyotime Biotech, Shanghai, China), were used. BCA protein assay kit and ROS assay kit were from the Beyotime Biotech. SH-SY5Y, N2a and HEK293 cells were from the China Center for Type Culture Collection (CCTCC). Wild type *Caenorhabditis elegans* strain N2 and the transgenic strain NL5901 [unc-54p:: α synuclein::YFP + unc-119(+)] were obtained from the Caenorhabditis Genetics Center (University of Minnesota, MN, USA). All other chemicals were of the highest grade available.

4.2. Fluorescence resonance energy transfer (FRET) assay

Intracellular α -syn aggregation was measured by a FRET-based assay using α -syn-CFP and α -syn-YFP as described [38], which provides a reproducible measure of intracellular aggregation [64]. Plasmids encoding the C-terminal fluorescent protein fusions, α -syn-CFP and α -syn-YFP, were constructed using standard protocols. For added fibril induced intracellular aggregation, as recipient cells, HEK293 cells were transfected with 150 ng of α -syn-CFP and 450 ng of α -syn-YFP plasmid constructs using Lipofectamine 2000 (Invitrogen). 24 h later, cells were seeded into a 96-well plate and allowed to recover overnight. α -syn fibrils were pretreated with or without copper/iron ions for 12 h, then applied to cells to a final concentration of 500 nM for 24 h to induce intracellular aggregation. For transcellular propagation experiments, recipient cells were transfected as described earlier; fibril-originated cells were transfected with 600 ng of synT plasmid constructs, a fusion of α -syn with a C-terminal truncated EGFP that greatly increases its intrinsic aggregation propensity [27]. After 24 h, both cells were seeded into 96-well plates at a 1:1 ratio and co-cultured for 48 h. FRET measurements were obtained using a Biotek SynergyH1 fluorescence plate reader (Winooski, VT), and normalized as the fluorescence of FRET/donor as

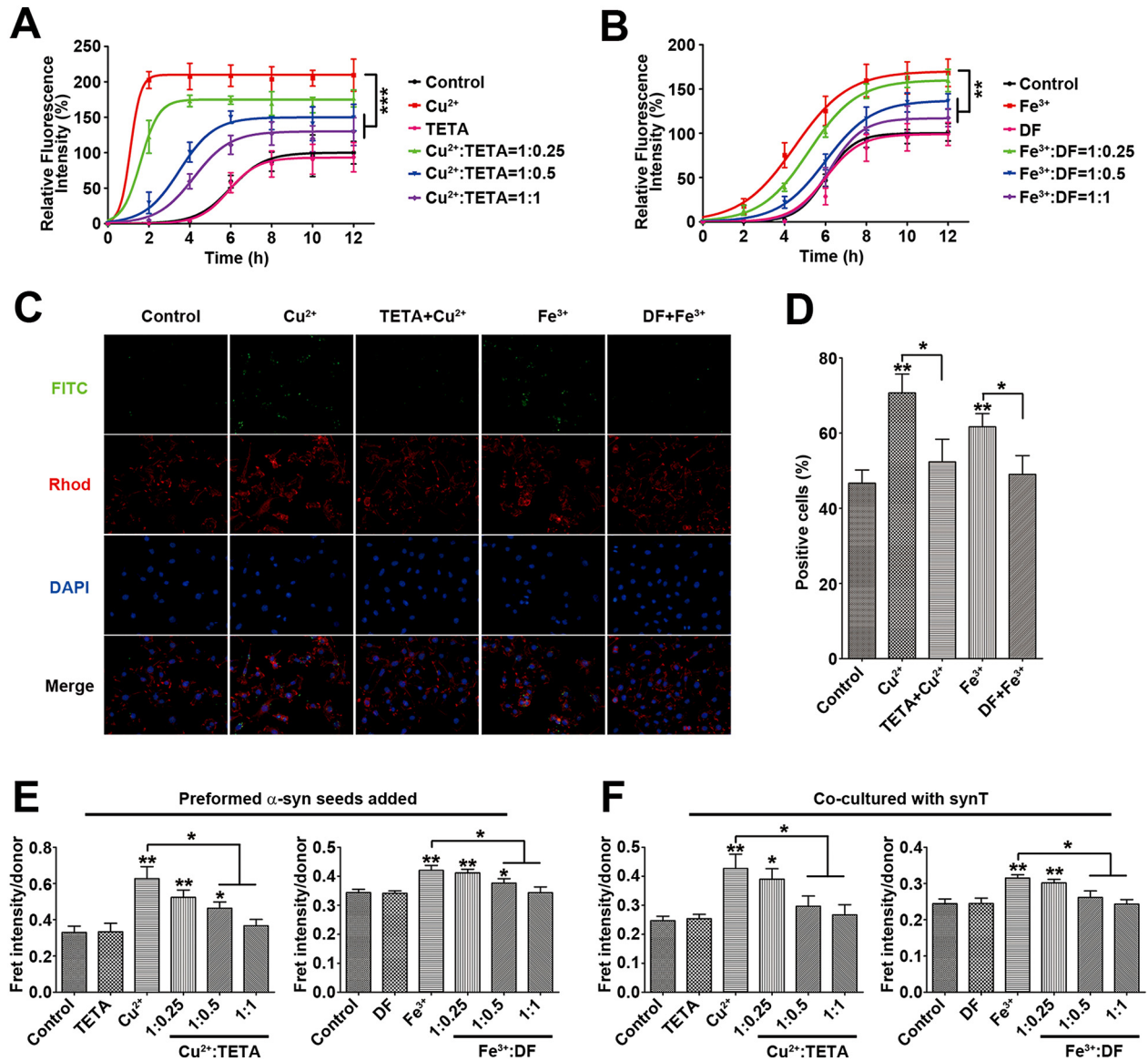


Fig. 6. Chelators abolish the effects of copper and iron ions on the seeded aggregation, fibril internalization and propagation. (A–B) TETA and DF dose-dependently reduced the facilitating effects of 100 μM copper (A) and iron (B) ions on seeded aggregation of α-syn. (C–D) TETA and DF (100 μM) abolished the promoting effects of copper/iron ions (100 μM) on fibril internalization in SH-SY5Y cells. The control was treated with α-syn-FITC fibrils only. All the images were taken at a magnification of 400×. (E–F) TETA and DF dose-dependently blocked the accelerating effects of 100 μM copper/iron ions on fibril induced intracellular α-syn-CFP/YFP aggregation (E) and transcellular fibril propagation (F). The control was treated with α-syn fibrils only. **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

Table 2
Amyloid fibril formation of seeded α-syn^a.

Chelators	Concentration (μM)	Metal ions	Concentration (μM)	Lag time (h)	Maximum intensity (%)
TETA	0	Cu ²⁺	0	4.33 ± 0.94	100.00 ± 15.86 ^b
	0		100	0.59 ± 0.23	209.48 ± 22.63 ^b
	100		0	4.57 ± 0.98	93.66 ± 20.71 ^b
	25		100	0.71 ± 0.29	176.69 ± 12.03 ^b
	50		100	1.80 ± 0.46	149.51 ± 18.66 ^b
	100		100	2.41 ± 0.69	129.79 ± 19.67 ^b
DF	0	Fe ³⁺	0	4.58 ± 0.99	100.00 ± 9.95 ^c
	0		100	1.87 ± 0.34	168.20 ± 15.51 ^c
	100		0	4.32 ± 0.88	99.22 ± 13.02 ^c
	25		100	2.85 ± 0.37	159.62 ± 12.60 ^c
	50		100	3.71 ± 0.82	135.98 ± 8.72 ^c
	100		100	4.58 ± 0.94	117.50 ± 10.14 ^c

^a α-syn concentration was 100 μM with 5 μM seeds.

^b The maximum intensity of the group without TETA and Cu²⁺ was set as 100%.

^c The maximum intensity of the group without DF and Fe³⁺ was set as 100%.

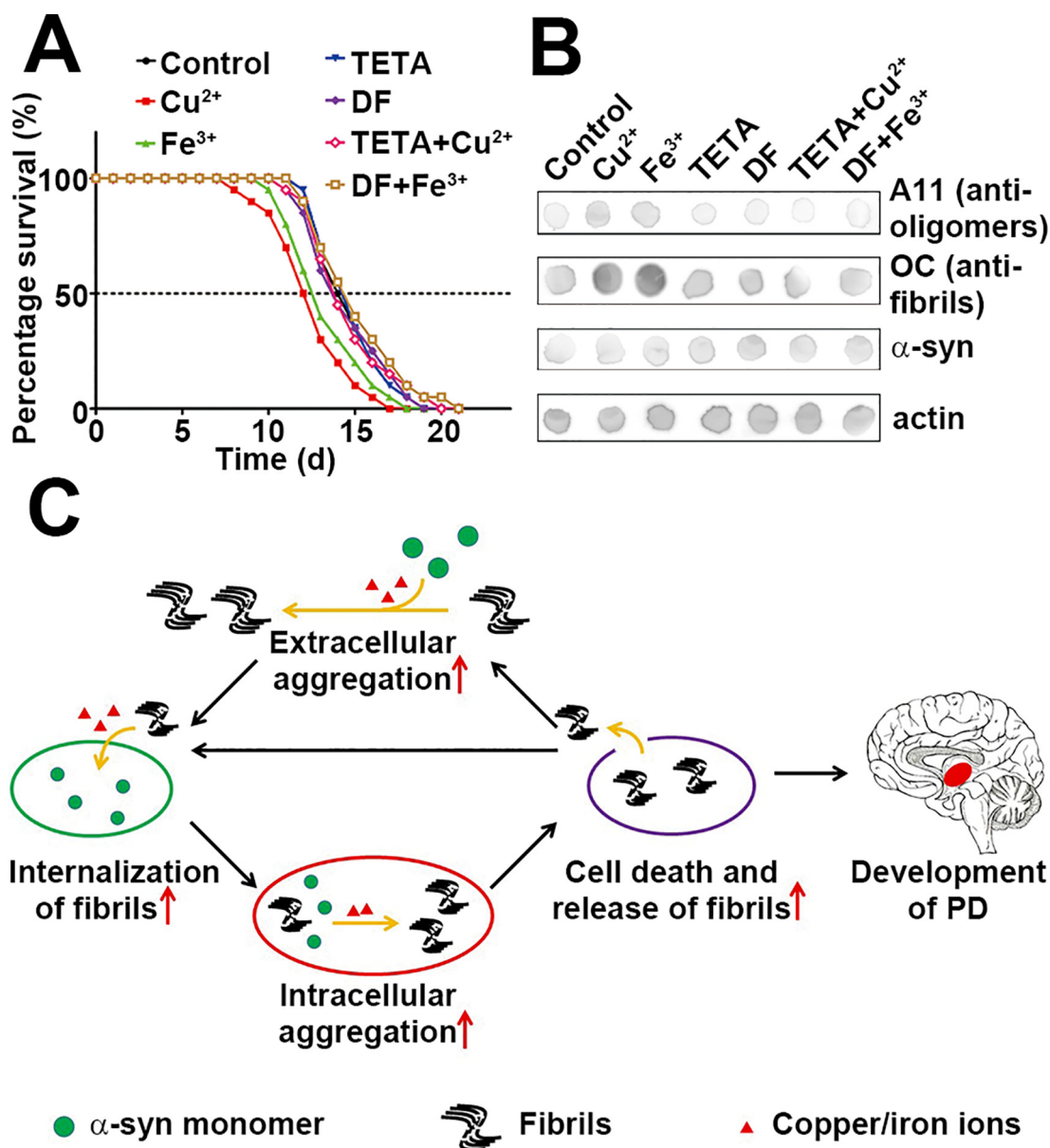


Fig. 7. Effects of copper/iron ions and chelators on *C. elegans*, and a working model for the role of copper/iron ions on α -syn fibrils propagation. (A) Effects of 1 μM copper/iron ions, chelators on the life span of *C. elegans* NL5901. (B) Immuno-dot-blot assays of α -syn oligomers, fibrils, total α -syn and actin of *C. elegans* NL5901 in the presence or absence of 1 μM chelators and metal ions identified by A11 (anti-oligomers), OC (anti-fibrils), anti- α -syn and anti-actin antibodies, respectively. (C) A schematic representation of the role of copper and iron ions on the propagation of α -syn fibrils.

described [38,64]. Excitation and emission wavelengths were CFP (435/485 nm excitation/emission), YFP (485/527 nm excitation/emission), and FRET (435/527 nm excitation/emission).

4.3. Amyloid formation and thioflavin-T (ThT) fluorescence assays

Amyloid formation using α -syn was performed as previously reported [65]. Briefly, α -syn was dissolved in 50 mM PBS buffer (containing 100 mM NaCl, pH 7.4) to a final concentration of 100 μM . The solution was sonicated for 2 min and centrifuged at 12,000g for 10 min to remove insoluble material. To form fibrils, 150 μL α -syn (100 μM) was incubated in a 96-well plate at 37 $^{\circ}\text{C}$ on a rotating shaker (220 rpm). For seeded aggregation, preformed fibrils were sonicated for 5 min and added as seeds (5%, final concentration 5 μM), in the presence or absence of copper/iron ions. ThT fluorescence assay and curve

fitting was performed as previously described [66,67]. Each experiment was repeated at least three times.

4.4. Transmission electron microscopy (TEM)

As previously described [68], 5 μL of the sample was dropped onto a 300-mesh formvar-carbon coated copper grid (Shanghai, China), followed by staining with 1% uranyl formate for 5 min, air drying, and analysis using an 80 kV H-8100 transmission electron microscope (Hitachi, Tokyo, Japan).

4.5. Far-UV circular dichroism (CD)

Far-UV CD was performed to study the solution secondary structure as previously described [69]. Samples were analyzed using a JASCO-810 circular dichroism spectropolarimeter (JASCO, Japan) under continuous

N₂ flow. The sample concentration was 10 µM and the CD spectra were obtained from 240 nm to 195 nm with a 2 nm band width, 100 nm/min scanning speed. Each result is given as the average of three experiments. The data obtained were converted into the mean residue ellipticity [θ] and analyzed with CDPro software using the CONTINLL algorithm [70].

4.6. Measurements of turbidity and soluble protein

Turbidity assay was performed using a Hitachi U-2900 spectrophotometer (Tokyo, Japan). Seeded aggregation of α-syn was performed as described above. Turbidity was measured at 400 nm for 30 s with a 2-h interval, and the average absorbance was recorded. For measurement of soluble protein, samples were removed at the indicated incubation time, centrifuged at 12,000g for 30 min to remove insoluble materials and the concentration of soluble protein was measured by the BCA assay.

4.7. Fourier Transform Infrared Spectrometry (FTIR)

As previously described [71], samples were incubated for 48 h, centrifuged at 12,000g for 30 min, the fibril pellets were washed 3 times, then lyophilized. Analysis was performed using a Fourier Transform Infrared Spectrometer (ATR-FTIR, Bruker Optics, Germany). Each spectrum contains 64 independent scans with a scan range of 4000–400 cm⁻¹ and a spectral resolution of 2 cm⁻¹. FTIR spectra corresponding to amide-I region (1700–1600 cm⁻¹) were analyzed for secondary structure. Spectra were normalized with the Origin 8, and curve fitting was based on the damped least-squares optimization algorithm assuming Gaussian band envelopes. Peaks in the curve fitted spectra were assigned in accordance with published reports [72,73].

4.8. Preparation of FITC labeled α-syn (α-syn-FITC) fibrils

α-Syn (100 µM) and FITC (500 µM) were co-incubated at 4 °C overnight, then 5 mM NH₄Cl was added to quench free FITC. Next, the solution was dialyzed against pure water twice and lyophilized [74]. Fibril formation of α-syn-FITC was performed as described earlier for 48 h, obtained fibrils were washed for 3 times, then lyophilized for further experiments.

4.9. Cell culture and analysis of fibril internalization

SH-SY5Y (a human neuroblastoma cell line) and Neuro-2a (N2a; a mouse neuroblastoma cell line) cells were cultured as described [50]. α-syn-FITC fibrils were pretreated with or without copper and iron ions for 12 h. After being exposed to 100 nM α-syn-FITC fibrils for 3 h, cells were washed twice with PBS, trypsinized to remove non-specifically bound α-syn-FITC fibrils, followed by plating on slides. Cells were allowed to recover for 4.5 h before fixation in 4% PFA, followed by staining with phalloidin and DAPI. To block heparan sulfate proteoglycan mediated internalization, both cells and α-syn-FITC fibrils were pretreated with 20 µg/mL heparin for 12 h before exposing cells to fibrils for 3 h. Images were taken using a CDD equipped fluorescence microscope (Sunny Optical, China). Quantification of images was conducted by using Image-Pro Plus digitizing software (IPP 6.0, Olympus).

4.10. Analysis of zeta potential

Zeta potential was analyzed using a zeta pals potential analyzer (Brookhaven Instruments, New York, NY). Briefly, 20 µM α-syn samples were sonicated for 30 min before the measurement. Each sample was scanned for 3 times (30 s per scan) and the mean zeta potential was recorded.

4.11. Knockdown of Ext1 in SH-SY5Y cells

Knockdown of *Ext1* was performed as previously described [75–77]. The shRNA (5' GCTTCAAAGTCTACGTATA 3') was constructed into pSUPER retro puro vector and then transfected into SH-SY5Y cells using Lipofectamine 2000. After 7-day selection with 2 µg/mL puromycin, knockdown efficacy was determined by Q-PCR.

4.12. Chemical modification of α-syn fibrils

Chemical modification of carboxyl was performed as previously described [78]. The carboxyl in mature α-syn fibrils was first modified by EDC and sulfo-NHS in MES buffer (100 mM, pH 5.0) to form sulfo-NHS ester, which then reacted with ethylamine in the sodium carbonate (150 mM, pH 8.5, Fig. S8D).

4.13. Analysis of α-syn aggregates in medium

Western blots were performed as previously described [51,52]. To determine whether intracellular α-syn aggregates were released to the extracellular space, culture medium was analyzed as described [24]. Briefly, HEK293 cells were transfected with Flag-α-syn, an N-terminal Flag-fused α-syn, in 12-well plate as acceptor cells. 12 h later, cells were treated with α-syn fibrils (500 nM) for 24 h, or co-cultured with equal number of cells expressing synT for 48 h, to induce intracellular Flag-α-syn aggregation. Then 1 mL medium was collected, concentrated, and the released Flag-α-syn aggregates were analyzed using western blots.

4.14. MTT assay

MTT-based toxicity assays were performed as previously described [79]. SH-SY5Y or N2a cells were plated in 96-well plates at a density of 5 × 10³ cells/well and cultured for 24 h. After addition of fresh medium containing α-syn samples, cells were incubated for another 24 h, 10 µL of MTT (5 mg/mL) and 100 µL DMSO were added to each well. The absorbance was measured at 570 nm.

4.15. ROS assay

ROS assay was performed as previously reported [80]. Briefly, cells were seeded in 24-well plates and exposed to one of the following conditions for 24 h: 10 µM α-syn, 10 µM α-syn fibrils, 10 µM Cu-fibrils, or 10 µM Fe-fibrils. Then cells were incubated in serum-free medium containing 10 µM DCFH-DA at 37 °C for 30 min, a positive control group was treated with 50 µg/mL ROSUP. ROS level was measured at excitation/emission wavelength of 488/525 nm. Absorbance was normalized to the respective control, which was set as one.

4.16. Analysis of *C. elegans* life span

As we previously described [63], 100 µL *E. coli* OP50 containing respective compound was added onto nematode growth medium (NGM) spread in a 35 mm dish. Transgenic *C. elegans* NL5901 (expressing a fusion of α-synuclein with yellow fluorescence protein, YFP) was synchronized with a bleach buffer (1 M NaOH, 1.2 M NaClO) for 3 min with gentle shaking. After three days, L3 larvae were washed with M9 buffer (22 mM KH₂PO₄, 42.3 mM Na₂HPO₄, 85.5 mM NaCl, 1 mM MgSO₄) and centrifuged (3.0 × 10³ rpm), and then transferred onto NGM. Each group (blank control, and Cu²⁺, Fe³⁺, TETA, DF, TETA with Cu²⁺, DF with Fe³⁺ treated groups) contains 140 larvae in 7 plates and incubated at 20 °C. *C. elegans* were transferred onto newly treated NGM plates every day, the number of *C. elegans* was counted under a ZSA 302 digital stereo microscope (COOC, Chongqing, China) and analyzed by using the GraphPad Prism.

4.17. Dot-blot assay of protein extracted from *C. elegans*

Synchronized L3 *C. elegans* were transferred into OP50 in 96-well plates, which contained the compound, at a density of about 20 larvae per well. After three days incubation at 20 °C, *C. elegans* were washed three times with M9 buffer and centrifuged. Then lysis buffer (50 mM Tris-HCl, 2 mM EDTA, 100 mM NaCl, 0.5% (v/v) Triton X-100, pH 8.5) was added and sonicated. The obtained solutions were quantified by using a BCA Protein Assay kit (Beyotime Biotechnology, Shanghai, China), and equal amounts of protein were blotted onto a nitrocellulose membrane for dot-blot assay as described previously [63,68].

4.18. Statistical analysis

Each experiment was repeated at least three times and the data were expressed as the average $\pm$ standard deviation (SD). The Kruskal-Wallis test and the Mann-Whitney test were used to evaluate statistical significance. Differences were considered statistically significant at $p < 0.05$.

Author contributions

K.H., Y.L. designed the research; Y.L., C.Y., S.W., D.Y., Y.Z., M.L., J.Z. performed experiments; Y.L., C.Y., H.C., R.B.P., L.X., and L.Z. analyzed data; Y.L., L.Z., and K.H. wrote the paper.

Declaration of competing interest

We declare that there are no conflicts of interest with this work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijbiomac.2020.06.274>.

References

- [1] A. Samii, J.G. Nutt, B.R. Ransom, Parkinson's disease, *Lancet* 363 (9423) (2004) 1783–1793.
- [2] B. Cheng, H. Gong, H. Xiao, R.B. Petersen, L. Zheng, K. Huang, Inhibiting toxic aggregation of amyloidogenic proteins: a therapeutic strategy for protein misfolding diseases, *Biochim. Biophys. Acta* 1830 (10) (2013) 4860–4871.
- [3] L. Huang, X. Liu, B. Cheng, K. Huang, How our bodies fight amyloidosis: effects of physiological factors on pathogenic aggregation of amyloidogenic proteins, *Arch. Biochem. Biophys.* 568 (2015) 46–55.
- [4] W.J. Schulz-Schaeffer, The synaptic pathology of alpha-synuclein aggregation in dementia with Lewy bodies, Parkinson's disease and Parkinson's disease dementia, *Acta Neuropathol.* 120 (2) (2010) 131–143.
- [5] M.G. Spillantini, M.L. Schmidt, V.M. Lee, J.Q. Trojanowski, R. Jakes, M. Goedert, Alpha-synuclein in Lewy bodies, *Nature* 388 (6645) (1997) 839–840.
- [6] N.M. Bonini, B.I. Giasson, Snaring the function of alpha-synuclein, *Cell* 123 (3) (2005) 359–361.
- [7] M.A. Alim, Q.L. Ma, K. Takeda, T. Aizawa, M. Matsubara, M. Nakamura, A. Asada, T. Saito, H. Kaji, M. Yoshii, S. Hisanaga, K. Ueda, Demonstration of a role for alpha-synuclein as a functional microtubule-associated protein, *J. Alzheimers Dis.* 6 (4) (2004) 435–442 (discussion 443–9).
- [8] S. Mohammadi, M. Nikkhah, S. Hosseinkhani, Loss in toxic function of aggregates of α -synuclein mutants by a β -synuclein derived peptide, *Protein Pept Lett* 24 (8) (2017) 757–764.
- [9] K.M. Danzer, L.R. Kranich, W.P. Ruf, O. Cagsal-Getkin, A.R. Winslow, L. Zhu, C.R. Vanderburg, P.J. McLean, Exosomal cell-to-cell transmission of alpha synuclein oligomers, *Mol. Neurodegener.* 7 (2012) 42.
- [10] G.K. Tofaris, M. Goedert, M.G. Spillantini, The transcellular propagation and intracellular trafficking of alpha-synuclein, *Cold Spring Harb Perspect Med* 7 (9) (2017) a024380.
- [11] E.J. Bae, N.Y. Yang, C. Lee, S. Kim, H.J. Lee, S.J. Lee, Haploinsufficiency of cathepsin D leads to lysosomal dysfunction and promotes cell-to-cell transmission of alpha-synuclein aggregates, *Cell Death Dis.* 6 (2015) e1901.
- [12] N. Van Den Berge, N. Ferreira, H. Gram, T.W. Mikkelsen, A.K.O. Alstrup, N. Casadei, P. Tsung-Pin, O. Riess, J.R. Nyengaard, G. Tamguney, P.H. Jensen, P. Borghammer, Evidence for bidirectional and trans-synaptic parasympathetic and sympathetic propagation of alpha-synuclein in rats, *Acta Neuropathol.* 138 (4) (2019) 535–550.
- [13] S. Kim, S.H. Kwon, T.I. Kam, N. Panicker, S.S. Karuppagounder, S. Lee, J.H. Lee, W.R. Kim, M. Kook, C.A. Foss, C. Shen, H. Lee, S. Kulkarni, P.J. Pasricha, G. Lee, M.G. Pomper, V.L. Dawson, T.M. Dawson, H.S. Ko, Transneuronal propagation of pathologic alpha-synuclein from the gut to the brain models Parkinson's disease, *Neuron* 103 (4) (2019) 627–641 e7.
- [14] A. Tarutani, T. Arai, S. Murayama, S.I. Hisanaga, M. Hasegawa, Potent prion-like behaviors of pathogenic alpha-synuclein and evaluation of inactivation methods, *Acta Neuropathol Commun* 6 (1) (2018) 29.
- [15] A. Okuzumi, M. Kurosawa, T. Hatano, M. Takanashi, S. Nojiri, T. Fukuhara, T. Yamanaka, H. Miyazaki, S. Yoshinaga, Y. Furukawa, T. Shimogori, N. Hattori, N. Nukina, Rapid dissemination of alpha-synuclein seeds through neural circuits in an in-vivo prion-like seeding experiment, *Acta Neuropathol Commun* 6 (1) (2018) 96.
- [16] Y. Chu, S. Muller, A. Tavares, O. Barret, D. Alagille, J. Seibyl, G. Tamagnan, K. Marek, K.C. Luk, J.Q. Trojanowski, V.M.Y. Lee, J.H. Kordower, Intrastriatal alpha-synuclein fibrils in monkeys: spreading, imaging and neuropathological changes, *Brain* 142 (11) (2019) 3565–3579.
- [17] V. Grodzanov, K.M. Danzer, Release and uptake of pathologic alpha-synuclein, *Cell Tissue Res.* 373 (1) (2018) 175–182.
- [18] T. Yasuda, Y. Nakata, H. Mochizuki, alpha-Synuclein and neuronal cell death, *Mol. Neurobiol.* 47 (2) (2013) 466–483.
- [19] P. Riederer, E. Sofic, W.D. Rausch, B. Schmidt, G.P. Reynolds, K. Jellinger, M.B. Youdim, Transition metals, ferritin, glutathione, and ascorbic acid in Parkinsonian brains, *J. Neurochem.* 52 (2) (1989) 515–520.
- [20] Y. Pushkar, G. Robison, B. Sullivan, S.X. Fu, M. Kohne, W. Jiang, S. Rohr, B. Lai, M.A. Marcus, T. Zakharova, W. Zheng, Aging results in copper accumulations in glial fibrillary acidic protein-positive cells in the subventricular zone, *Aging Cell* 12 (5) (2013) 823–832.
- [21] A. Reeve, E. Simcox, D. Turnbull, Ageing and Parkinson's disease: why is advancing age the biggest risk factor? *Ageing Res. Rev.* 14 (2014) 19–30.
- [22] H.S. Pall, A.C. Williams, D.R. Blake, J. Lunec, J.M. Gutteridge, M. Hall, A. Taylor, Raised cerebrospinal-fluid copper concentration in Parkinson's disease, *Lancet* 2 (8553) (1987) 238–241.
- [23] S. Mohammadi, M. Nikkhah, S. Hosseinkhani, Investigation of the effects of carbon-based nanomaterials on A53T alpha-synuclein aggregation using a whole-cell recombinant biosensor, *Int. J. Nanomedicine* 12 (2017) 8831–8840.
- [24] N. Kfoury, B.B. Holmes, H. Jiang, D.M. Holtzman, M.I. Diamond, Trans-cellular propagation of Tau aggregation by fibrillar species, *J. Biol. Chem.* 287 (23) (2012) 19440–19451.
- [25] T.R. Yamasaki, B.B. Holmes, J.L. Furman, D.D. Dhavale, B.W. Su, E.S. Song, N.J. Cairns, P.T. Kotzbauer, M.I. Diamond, Parkinson's disease and multiple system atrophy have distinct alpha-synuclein seed characteristics, *J. Biol. Chem.* 294 (3) (2019) 1045–1058.
- [26] N. Rezaei, N. Shirvanizadeh, S. Mohammadi, M. Nikkhah, S.S. Arab, The inhibitory effects of biomimetically designed peptides on alpha-synuclein aggregation, *Arch. Biochem. Biophys.* 634 (2017) 96–106.
- [27] P.J. McLean, H. Kawamata, B.T. Hyman, Alpha-synuclein-enhanced green fluorescent protein fusion proteins form proteasome sensitive inclusions in primary neurons, *Neuroscience* 104 (3) (2001) 901–912.
- [28] J. Pujols, S. Pena-Diaz, D.F. Lázaro, F. Peccati, F. Pinheiro, D. Gonzalez, A. Carija, S. Navarro, M. Conde-Gimenez, J. Garcia, S. Guardiola, E. Giral, X. Salvatella, J. Sancho, M. Sodupe, T.F. Outeiro, E. Dalfó, S. Ventura, Small molecule inhibits alpha-synuclein aggregation, disrupts amyloid fibrils, and prevents degeneration of dopaminergic neurons, *Proc. Natl. Acad. Sci. U. S. A.* 115 (41) (2018) 10481–10486.
- [29] V.N. Uversky, J. Li, A.L. Fink, Metal-triggered structural transformations, aggregation, and fibrillation of human alpha-synuclein. A possible molecular link between Parkinson's disease and heavy metal exposure, *J. Biol. Chem.* 276 (47) (2001) 44284–44296.
- [30] A. Lau, R.W.L. So, H.H.C. Lau, J.C. Sang, A. Ruiz-Riquelme, S.C. Fleck, E. Stuart, S. Menon, N.P. Visanji, G. Meisl, R. Faidi, M.M. Marano, C. Schmitt-Ulms, Z. Wang, P.E. Fraser, A. Tandon, B.T. Hyman, H. Wille, M. Ingelsson, D. Klenerman, J.C. Watts, alpha-Synuclein strains target distinct brain regions and cell types, *Nat. Neurosci.* 23 (1) (2020) 21–31.
- [31] N.L. Rey, L. Bousset, S. George, Z. Madaj, L. Meyerdirk, E. Schulz, J.A. Steiner, R. Melki, P. Brundin, alpha-Synuclein conformational strains spread, seed and target neuronal cells differentially after injection into the olfactory bulb, *Acta Neuropathol Commun* 7 (1) (2019) 221.
- [32] S. Griboaud, P. Tixador, L. Bousset, A. Fenyi, P. Lino, R. Melki, J.M. Peyrin, A.L. Perrier, Propagation of alpha-synuclein strains within human reconstructed neuronal network, *Stem Cell Reports* 12 (2) (2019) 230–244.

- [33] M.S. Planchard, S.E. Exley, S.E. Morgan, V. Rangachari, Dopamine-induced alpha-synuclein oligomers show self- and cross-propagation properties, *Protein Sci.* 23 (10) (2014) 1369–1379.
- [34] A. Tarutani, G. Suzuki, A. Shimozawa, T. Nonaka, H. Akiyama, S. Hisanaga, M. Hasegawa, The effect of fragmented pathogenic alpha-synuclein seeds on prion-like propagation, *J. Biol. Chem.* 291 (36) (2016) 18675–18688.
- [35] J.M. Froula, M. Castellana-Cruz, N.M. Anabtawi, J.D. Camino, S.W. Chen, D.R. Thrasher, J. Freire, A.A. Yazdi, S. Fleming, C.M. Dobson, J.R. Kumita, N. Cremades, L.A. Volpicelli-Daley, Defining alpha-synuclein species responsible for Parkinson's disease phenotypes in mice, *J. Biol. Chem.* 294 (27) (2019) 10392–10406.
- [36] H. Hayakawa, R. Nakatani, K. Ikenaka, C. Aguirre, C.J. Choong, H. Tsuda, S. Nagano, M. Koike, T. Ikeuchi, M. Hasegawa, S.M. Papa, Y. Nagai, H. Mochizuki, K. Baba, Structurally distinct alpha-synuclein fibrils induce robust parkinsonian pathology, *Mov. Disord.* 35 (2) (2020) 256–267.
- [37] S. Sarrazin, W.C. Lamanna, J.D. Esko, Heparan sulfate proteoglycans, *Cold Spring Harb. Perspect. Biol.* 3 (7) (2011) (pii: a004952).
- [38] B.B. Holmes, S.L. DeVos, N. Kfoury, M. Li, R. Jacks, K. Yanamandra, M.O. Ouidja, F.M. Brodsky, J. Marasa, D.P. Bagchi, P.T. Kotzbauer, T.M. Miller, D. Papy-Garcia, M.I. Diamond, Heparan sulfate proteoglycans mediate internalization and propagation of specific proteopathic seeds, *Proc. Natl. Acad. Sci. U. S. A.* 110 (33) (2013) E3138–E3147.
- [39] X. Zhang, E. Wesen, R. Kumar, D. Bernson, A. Gallud, A. Paul, P. Wittung-Stafshede, E.K. Esbjörner, Correlation between cellular uptake and cytotoxicity of fragmented alpha-synuclein amyloid fibrils suggests intracellular basis for toxicity, *ACS Chem. Neurosci.* 11 (3) (2020) 233–241.
- [40] B.E. Stopschinski, B.B. Holmes, G.M. Miller, V.A. Manon, J. Vaquer-Alicea, W.L. Prueitt, L.C. Hsieh-Wilson, M.I. Diamond, Specific glycosaminoglycan chain length and sulfation patterns are required for cell uptake of tau versus alpha-synuclein and beta-amyloid aggregates, *J. Biol. Chem.* 293 (27) (2018) 10826–10840.
- [41] K. Lidholt, J.L. Weinke, C.S. Kiser, F.N. Lugenma, K.J. Bame, S. Cheifetz, J. Massague, U. Lindahl, J.D. Esko, A single mutation affects both N-acetylglucosaminyltransferase and glucuronosyltransferase activities in a Chinese hamster ovary cell mutant defective in heparan sulfate biosynthesis, *Proc. Natl. Acad. Sci. U. S. A.* 89 (6) (1992) 2267–2271.
- [42] F. Lermyte, J. Everett, J. Brooks, F. Bellingeri, K. Billimoria, P.J. Sadler, P.B. O'Connor, N.D. Telling, J.F. Collingwood, Emerging approaches to investigate the influence of transition metals in the proteinopathies, *Cells* 8 (10) (2019) (pii: E1231).
- [43] Y. Tian, H.F. Stanyon, J.D. Barritt, U. Mayet, P. Patel, E. Karamani, G. Fusco, J.H. Viles, Copper(2+) binding to alpha-synuclein. Histidine50 can form a ternary complex with Cu(2+) at the N-terminus but not a macrochelate, *Inorg. Chem.* 58 (22) (2019) 15580–15589.
- [44] J. Han, H.J. Lee, K.Y. Kim, G. Nam, J. Chae, M.H. Lim, Mechanistic approaches for chemically modifying the coordination sphere of copper-amyloid-beta complexes, *Proc. Natl. Acad. Sci. U. S. A.* 117 (10) (2020) 5160–5167.
- [45] C. Wang, C. Yang, Y.C. Chen, L. Ma, K. Huang, Rational design of hybrid peptides: a novel drug design approach, *Curr Med Sci* 39 (3) (2019) 349–355.
- [46] H. Inouye, K.A. Gleason, D. Zhang, S.M. Decatur, D.A. Kirschner, Differential effects of Phe19 and Phe20 on fibril formation by amyloidogenic peptide A beta 16–22 (Ac-KLVFFAE-NH₂), *Proteins* 78 (10) (2010) 2306–2321.
- [47] Y.Q. Wang, M. Bongiovanni, S.L. Gras, S. Perrett, The fibrils of Ure2p homologs from *Saccharomyces cerevisiae* and *Saccharomyces paradoxus* have similar cross-beta structure in both dried and hydrated forms, *J. Struct. Biol.* 174 (3) (2011) 505–511.
- [48] M. Brahic, L. Bousset, G. Bieri, R. Melki, A.D. Gitler, Axonal transport and secretion of fibrillar forms of alpha-synuclein, A beta42 peptide and HTTexon 1, *Acta Neuropathol.* 131 (4) (2016) 539–548.
- [49] P. Cao, A. Abedini, H. Wang, L.H. Tu, X. Zhang, A.M. Schmidt, D.P. Raleigh, Islet amyloid polypeptide toxicity and membrane interactions, *Proc. Natl. Acad. Sci. U. S. A.* 110 (48) (2013) 19279–19284.
- [50] L. Ma, C. Yang, X. Zhang, Y. Li, S. Wang, L. Zheng, K. Huang, C-terminal truncation exacerbates the aggregation and cytotoxicity of alpha-synuclein: a vicious cycle in Parkinson's disease, *Biochim. Biophys. Acta Mol. Basis Dis.* 1864 (12) (2018) 3714–3725.
- [51] W. Wang, Q. Wang, D. Wan, Y. Sun, L. Wang, H. Chen, C. Liu, R.B. Petersen, J. Li, W. Xue, L. Zheng, K. Huang, Histone H1T1C/H1.2 regulates autophagy in the development of diabetic retinopathy, *Autophagy* 13 (5) (2017) 941–954.
- [52] H. Chen, L. Wang, W. Wang, C. Cheng, Y. Zhang, Y. Zhou, C. Wang, X. Miao, J. Wang, C. Wang, J. Li, L. Zheng, K. Huang, ELABELA and an ELABELA fragment protect against AKI, *J. Am. Soc. Nephrol.* 28 (9) (2017) 2694–2707.
- [53] N.K. Polinski, L.A. Volpicelli-Daley, C.E. Sortwell, K.C. Luk, N. Cremades, L.M. Gottler, J. Froula, M.F. Duffy, V.M.Y. Lee, T.N. Martinez, K.D. Dave, Best practices for generating and using alpha-synuclein pre-formed fibrils to model Parkinson's disease in rodents, *J. Park. Dis.* 8 (2) (2018) 303–322.
- [54] B. Winner, R. Jappelli, S.K. Maji, P.A. Desplats, L. Boyer, S. Aigner, C. Hetzer, T. Lohr, M. Vilar, S. Campioni, C. Tzitzilonis, A. Soragni, S. Jessberger, H. Mira, A. Consiglio, E. Pham, E. Masliah, F.H. Gage, R. Riek, In vivo demonstration that alpha-synuclein oligomers are toxic, *Proc. Natl. Acad. Sci. U. S. A.* 108 (10) (2011) 4194–4199.
- [55] C. Appenzeller-Herzog, T. Mathes, M.L.S. Heeres, K.H. Weiss, R.H.J. Houwen, H. Ewald, Comparative effectiveness of common therapies for Wilson disease: a systematic review and meta-analysis of controlled studies, *Liver Int.* 39 (11) (2019) 2136–2152.
- [56] R.C. Hider, A.V. Hoffbrand, The role of deferiprone in iron chelation, *N. Engl. J. Med.* 379 (22) (2018) 2140–2150.
- [57] I. Hozumi, T. Hasegawa, A. Honda, K. Ozawa, Y. Hayashi, K. Hashimoto, M. Yamada, A. Koumura, T. Sakurai, A. Kimura, Y. Tanaka, M. Satoh, T. Inuzuka, Patterns of levels of biological metals in CSF differ among neurodegenerative diseases, *J. Neurol. Sci.* 303 (1–2) (2011) 95–99.
- [58] L. Ma, C. Yang, J. Zheng, Y. Chen, Y. Xiao, K. Huang, Non-polyphenolic natural inhibitors of amyloid aggregation, *Eur. J. Med. Chem.* 192 (2020), 112197.
- [59] S. Bolognin, L. Messori, D. Drago, C. Gabbiani, L. Cendron, P. Zatta, Aluminum, copper, iron and zinc differentially alter amyloid-A beta(1–42) aggregation and toxicity, *Int. J. Biochem. Cell Biol.* 43 (6) (2011) 877–885.
- [60] Y. Liu, M. Nguyen, A. Robert, B. Meunier, Metal ions in Alzheimer's disease: a key role or not? *Acc. Chem. Res.* 52 (7) (2019) 2026–2035.
- [61] A. Gaeta, R.C. Hider, The crucial role of metal ions in neurodegeneration: the basis for a promising therapeutic strategy, *Br. J. Pharmacol.* 146 (8) (2005) 1041–1059.
- [62] Y. Li, Z. Wang, Y. Chen, R.B. Petersen, L. Zheng, K. Huang, Salvation of the fallen angel: reactivating mutant p53, *Br. J. Pharmacol.* 176 (7) (2019) 817–831.
- [63] K. Ji, Y. Zhao, T. Yu, Z. Wang, H. Gong, X. Yang, Y. Liu, K. Huang, Inhibition effects of tanshinone on the aggregation of alpha-synuclein, *Food Funct.* 7 (1) (2016) 409–416.
- [64] S.K. Pollitt, J. Pallos, J. Shao, U.A. Desai, A.A. Ma, L.M. Thompson, J.L. Marsh, M.I. Diamond, A rapid cellular FRET assay of polyglutamine aggregation identifies a novel inhibitor, *Neuron* 40 (4) (2003) 685–694.
- [65] L. Giehm, D.E. Otzen, Strategies to increase the reproducibility of protein fibrillation in plate reader assays, *Anal. Biochem.* 400 (2) (2010) 270–281.
- [66] C. Guo, L. Ma, Y. Zhao, A. Peng, B. Cheng, Q. Zhou, L. Zheng, K. Huang, Inhibitory effects of magnolol and honokiol on human calcitonin aggregation, *Sci. Rep.* 5 (2015) 13556.
- [67] H. Gong, X. Zhang, B. Cheng, Y. Sun, C. Li, T. Li, L. Zheng, K. Huang, Bisphenol A accelerates toxic amyloid formation of human islet amyloid polypeptide: a possible link between bisphenol A exposure and type 2 diabetes, *PLoS One* 8 (1) (2013) e54198.
- [68] Y. Zhang, Y. Zhao, Z. Wang, H. Gong, L. Ma, D. Sun, C. Yang, Y. Li, B. Cheng, R.B. Petersen, F. Jiang, G. Liu, K. Huang, Menadione sodium bisulfite inhibits the toxic aggregation of amyloid-beta(1–42), *Biochim. Biophys. Acta Gen. Subj.* 1862 (10) (2018) 2226–2235.
- [69] L. Huang, X. Yang, A. Peng, H. Wang, X. Lei, L. Zheng, K. Huang, Inhibitory effect of leonurine on the formation of advanced glycation end products, *Food Funct.* 6 (2) (2015) 584–589.
- [70] L. Ma, C. Yang, L. Huang, Y. Chen, Y. Li, C. Cheng, B. Cheng, L. Zheng, K. Huang, Glycated insulin exacerbates the cytotoxicity of human islet amyloid polypeptides: a vicious cycle in type 2 diabetes, *ACS Chem. Biol.* 14 (3) (2019) 486–496.
- [71] Y. Li, L. Huang, X. Yang, C. Wang, Y. Sun, H. Gong, Y. Liu, L. Zheng, K. Huang, The effect of exposing a critical hydrophobic patch on amyloidogenicity and fibril structure of insulin, *Biochem. Biophys. Res. Commun.* 440 (1) (2013) 56–61.
- [72] K. Huang, N.C. Maiti, N.B. Phillips, P.R. Carey, M.A. Weiss, Structure-specific effects of protein topology on cross-beta assembly: studies of insulin fibrillation, *Biochemistry* 45 (34) (2006) 10278–10293.
- [73] Y. Li, J. Maurer, A. Roth, V. Vogel, E. Winter, W. Mantele, A setup for simultaneous measurement of infrared spectra and light scattering signals: watching amyloid fibrils grow from intact proteins, *Rev Sci Instrum* 85 (8) (2014), 084302.
- [74] Y. Zhang, W. Xue, W. Zhang, Y. Yuan, X. Zhu, Q. Wang, Y. Wei, D. Yang, C. Yang, Y. Chen, Y. Sun, S. Wang, K. Huang, L. Zheng, Histone methyltransferase G9a protects against acute liver injury through GSTP1, *Cell Death Differ.* 27 (4) (2020) 1243–1258.
- [75] S. Liu, Y. Sun, M. Jiang, Y. Li, Y. Tian, W. Xue, N. Ding, Y. Sun, C. Cheng, J. Li, X. Miao, X. Liu, L. Zheng, K. Huang, Glyceraldehyde-3-phosphate dehydrogenase promotes liver tumorigenesis by modulating phosphoglycerate dehydrogenase, *Hepatology* 66 (2) (2017) 631–645.
- [76] C. Liu, J. Wang, Y. Wei, W. Zhang, M. Geng, Y. Yuan, Y. Chen, Y. Sun, H. Chen, Y. Zhang, M. Xiong, Y. Li, L. Zheng, K. Huang, Fat-specific knockout of Mecn2 upregulates S1p1 to reduce obesity by enhancing browning, *Diabetes* 69 (1) (2020) 35–47.
- [77] Y. Sun, M. Geng, Y. Yuan, P. Guo, Y. Chen, D. Yang, R.B. Petersen, K. Huang, L. Zheng, Lmo4-resistin signaling contributes to adipose tissue-liver crosstalk upon weight cycling, *FASEB J.* 34 (3) (2020) 4732–4748.
- [78] Z. Grabarek, J. Gergely, Zero-length crosslinking procedure with the use of active esters, *Anal. Biochem.* 185 (1) (1990) 131–135.
- [79] L. Jiao, X. Zhang, L. Huang, H. Gong, B. Cheng, Y. Sun, Y. Li, Q. Liu, L. Zheng, K. Huang, Proanthocyanidins are the major anti-diabetic components of cinnamon water extract, *Food Chem. Toxicol.* 56 (2013) 398–405.
- [80] R. Cai, W. Xue, S. Liu, R.B. Petersen, K. Huang, L. Zheng, Overexpression of glyceraldehyde 3-phosphate dehydrogenase prevents neurovascular degeneration after retinal injury, *FASEB J.* 29 (7) (2015) 2749–2758.